Homework #1

This is the homework for week 2. The due date is 11:59 p.m., September 22. Late submission will automatically result in "0" grade. Please include codes for the program part in your submission.

1 Computation Questions

(**Question1):** (30 points) i. Suppose random variable X satisfies Possion distribution with parameter $\lambda > 0$, i.e., the probability mass function

$$Prob(X = k) = \frac{\lambda^k}{k!}\exp(-\lambda), \text{ where } k = 0, 1, \cdots,$$

Calculate $\mathbf{E}X$, $Var(X)$, and its moment generating function $H(t) = \mathbf{E}\exp(tX)$, where $t \in \mathbf{R}$.

ii. Denote $F(a)$ to be the cumulative distribution function of X, prove that $F(a) \geq 1 - \frac{\lambda}{a}$ for any $a > 0$.

iii. Now suppose random variable Y satisfies uniform distribution on $[0, 1]$, i.e., its density $f(t) = 1$ on $[0, 1]$. Derive the mean $\mathbf{E}Y$, the variance $Var(Y)$, the skewness γ, and the Kurtosis $Kurt(Y)$.

iv. Suppose $Z_1, Z_2, \cdots, Z_n$ are i.i.d. (independent and identical distributed) random variables, prove that

$$Var(\sum_{i=1}^{n} Z_i) = \sum_{i=1}^{n} Var(Z_i).$$

Hint: you can use this fact when you prove this question: If $Z_1, Z_2, \cdots, Z_n$ are independent, then for any function h, $h(Z_1, \cdots, Z_k)$ and Z_{k+1} is also independent.

Answers:

- i. By definition,

$$\mathbf{E}X = \sum_{k=1}^{\infty} k \times \frac{\lambda^k}{k!}\exp(-\lambda) = \lambda\sum_{k=1}^{\infty}\frac{\lambda^{k-1}}{(k-1)!}\exp(-\lambda) = \lambda,$$

and

$$\mathbf{E}X^2 = \sum_{k=2}^{\infty} k(k-1) \times \frac{\lambda^k}{k!}\exp(-\lambda) + \sum_{k=1}^{\infty} k \times \frac{\lambda^k}{k!}\exp(-\lambda)$$
$$= \lambda^2\sum_{k=2}^{\infty}\frac{\lambda^{k-2}}{(k-2)!}\exp(-\lambda) + \lambda\sum_{k=1}^{\infty}\frac{\lambda^{k-1}}{(k-1)!}\exp(-\lambda) = \lambda^2 + \lambda.$$

Therefore,

$$Var(X) = \mathbf{E}X^2 - (\mathbf{E}X)^2 = \lambda.$$

Moreover,

$$H(t) = \sum_{k=1}^{\infty}\exp(tk) \times \frac{\lambda^k}{k!}\exp(-\lambda) = \sum_{k=1}^{\infty}\frac{(\lambda\exp(t))^k}{k!}\exp(-\lambda) = \exp\left(\lambda\exp(t) - \lambda\right).$$

- From Markov inequality,

$$Prob(X > a) \leq \frac{\mathbf{E}X}{a} = \frac{\lambda}{a}, \text{ so } F(a) = Prob(X \leq a) \geq 1 - \frac{\lambda}{a}.$$

- The mean is

$$\mathbf{E}Y = \int_0^1 t\mathrm{d}t = \frac{1}{2}, \ \mathbf{E}Y^2 = \int_0^1 t^2\mathrm{d}t = \frac{1}{3}, \ \mathbf{E}Y^3 = \int_0^1 t^3\mathrm{d}t = \frac{1}{4}, \text{ and } \mathbf{E}Y^4 = 1/5.$$

Therefore,

$$\begin{aligned} Var(Y) = \mathbf{E}Y^2 - (\mathbf{E}Y)^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}, \\ \gamma = (1/4 - 3 \times \frac{1}{2} \times 1/3 + 3 \times \frac{1}{4} \times \frac{1}{2} - \frac{1}{8})/(1/12)^{3/2} = 0, \\ Kurt(Y) = -3 + \frac{\frac{1}{5} - 4 \times \frac{1}{2} \times \frac{1}{4} + 6 \times \frac{1}{4} \times \frac{1}{3} - 4 \times \frac{1}{8} \times \frac{1}{2} + \frac{1}{16}}{(1/12)^2} = -\frac{6}{5}. \end{aligned}$$

- From Induction,

$$Var(\sum_{i=1}^{n} Z_i) = Var(Z_n) + Var(\sum_{i=1}^{n-1} Z_i) = \sum_{i=1}^{n} Var(Z_i).$$

(**Question2):** (20 points)

(Hint, see problem 5.3–8. in the textbook) Suppose two independent claims X and Y are made on two insured homes, where each claim has probability density function

$$f(x) = \frac{4}{x^5}, 1 < x.$$

Find the expected value and the variance of the larger claim $\max(X, Y)$.

For the cdf of the maximum is

$$Prob(\max(X, Y) \leq x) = Prob(X \leq x) \times Prob(Y \leq x) = (1 - x^{-4})^2,$$

the density is

$$g(x) = 2(1 - x^{-4}) \times \frac{4}{x^5}.$$

The expectation is

$$\begin{aligned} \mathbf{E}\max(X, Y) = \int_1^\infty 2(1 - x^{-4}) \times \frac{4}{x^4} dx = \frac{32}{21}, \\ \text{and } \mathbf{E}\max(X, Y)^2 = \int_1^\infty 2(1 - x^{-4}) \times \frac{4}{x^2} dx = \frac{32}{5}. \end{aligned}$$

Therefore,

$$Var(\max(X, Y)) = \frac{32}{5} - (\frac{32}{21})^2 = 4.078.$$

(**Question3):** (30 points)

Suppose X and Y are independent random variables with mean $\mathbf{E}X = \mathbf{E}Y = \mu \neq 0$ and variance $Var(X) = Var(Y) = \sigma^2 > 0$. Find the correlation coefficient

$$Corr(XY, Y).$$

Notice that

$$\mathbf{E}XY \times Y = \mathbf{E}X \times \mathbf{E}Y^2 = \mu(\mu^2 + \sigma^2) \Rightarrow Cov(XY, Y) = \mu\sigma^2.$$

Therefore,

$$Corr(XY, Y) = \frac{\mu\sigma^2}{\sigma\sqrt{Var(XY)}} = \frac{\sigma^2\mu}{\sigma^2\sqrt{\sigma^2 + 2\mu^2}} = \frac{\mu}{\sqrt{\sigma^2 + 2\mu^2}}.$$

2 Programming Questions

(20 points)

Implement the following experiment using Jupyter or R:

i. Generate 500 i.i.d. random variables $X_1, \cdots, X_{500}$ with uniform distribution on $[0, 1]$.

ii. Calculate and record the sample mean, sample variace, sample skewness, and sample Kurtosis (See the "Estimator" column of Table 1 of the lecture notes "probability review").

iii. Repeat experiment for 300 times, plot four histograms of the sample values. Make comparisons between the sample values and the true values.